

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Pharmacy

Semester: IV

Subject Name: Physical Pharmaceutics-II

Subject Code: BP403T

Max Marks: 75

Date: 24/12/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All questions are compulsory.
2. Draw diagrams / figures wherever necessary.
3. Figures to right indicate full marks..

Q.1. Objective Type Questions (Answer all the questions)

(02 X 10) = 20

- i) Differentiate between elastic and plastic deformation.
- ii) Draw a well labelled sketch of Ostwald's viscometer.
- iii) Define second order reaction with suitable example.
- iv) What is gold number? Give examples.
- v) Enlist the various identification tests of emulsion.
- vi) How will you apply DLVO theory to stabilize the colloidal dispersion?
- vii) State the Newton's law of flow with equation.
- viii) Define flow activators. Give examples.
- ix) How will you compare flocculated and deflocculated suspensions?
- x) What is plug flow? How it can be minimized?

Q.2. Long Answers (Answer 2 out of 3)

(10 X 02) = 20

- i) Define micromeritics and particle size. Explain in detail the derived properties of powder.
- ii) What is first order of reaction? Derive an equation for determination of rate constant of a first order reaction. Discuss the methods for determination of order of reaction.
- iii) Define dispersed system. Describe in detail the different purification methods of colloids.

Q.3. Short Answers (Answer 7 out of 9)

(05 X 07) = 35

- i) Define angle of repose. Explain the fixed funnel method for determination of angle of repose.
- ii) What is colloid? How will you give the types of colloids according to the affinity towards solvent?
- iii) Describe optical microscopy method for particle size determination with its merits and demerits.
- iv) Discuss the pharmaceutical and biological applications of rheology.
- v) Explain electrical double layer theory with well labelled sketch.
- vi) What is phase inversion? How the different factors responsible for it.
- vii) Describe the various factors affecting the chemical degradation of pharmaceutical product.
- viii) Discuss any two types of Non-Newtonian flow with its mechanism and graph.
- ix) Define emulsion. Write the various instabilities in emulsion.

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